

Chapter 8: Human Physiology

Circulatory, Respiratory, Nervous, Endocrine & Immune Systems

8.1 Circulatory & Respiratory Systems

The cardiovascular system consists of a four-chambered heart, blood vessels, and blood. Oxygenated and deoxygenated blood flow through systemic and pulmonary circuits. The respiratory system facilitates gas exchange across the alveolar-capillary membrane in the lungs.

8.2 Nervous & Endocrine Systems

- **Nervous System:** Central (brain and spinal cord) and peripheral systems. Neurons transmit electrochemical signals via action potentials across synapses using neurotransmitters.
- **Endocrine System:** Secretes hormones directly into the bloodstream to regulate metabolism, growth, and homeostasis (e.g., insulin, cortisol, thyroxine).

8.3 Immune Defense & Homeostasis

The human body maintains homeostasis through feedback loops. The immune system features innate defenses (barriers, phagocytes) and adaptive immunity (B cells, T cells, and antibody-mediated antigen neutralization).